Do Now · retrieval

Lesson 108 · Interrupts and the Interrupt Service Routine

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

With page size 1,024 bytes, what offset is address 2,500?

Q2

For the binary value 1010, what is its decimal value?

Q3

A linked-list node has data and what other essential field?

Q4

In a BST, where is a value less than the current node placed?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Event occurs — A device or timer signals that attention is needed.

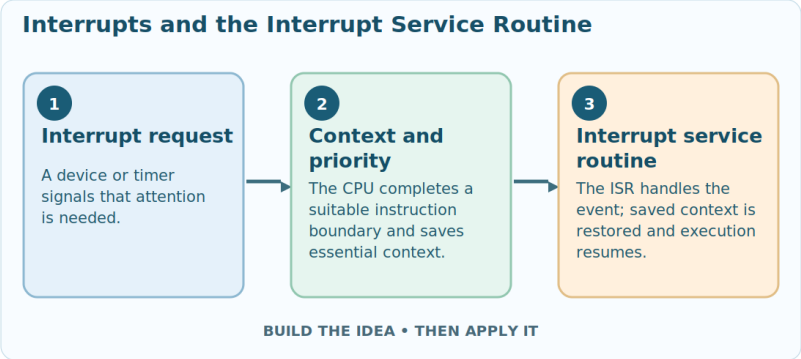
Core idea 2

State is preserved — The CPU completes a suitable instruction boundary and saves essential context.

Core idea 3

Service and return — The ISR handles the event; saved context is restored and execution resumes.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

interrupt — Signal requiring CPU attention.

ISR — Routine that services a particular interrupt.

context — CPU state needed to resume execution.

polling — Repeated checking of a device or condition.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

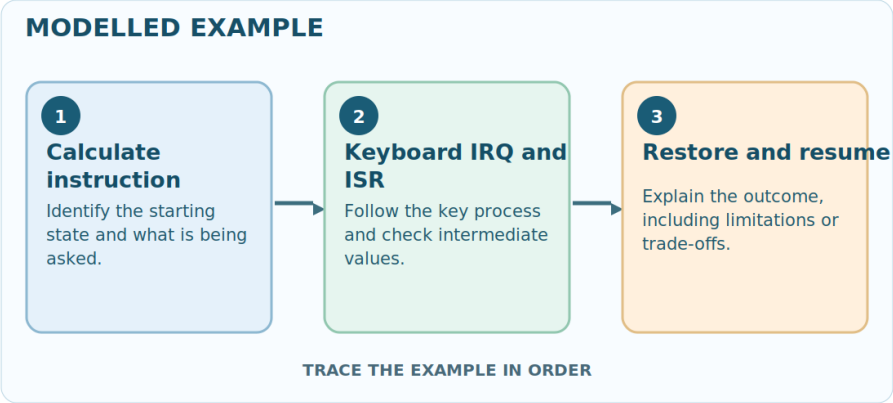
Problem

A keyboard interrupt arrives while a calculation is running. Trace the CPU response.

Follow the worked method

1. Model answer — After completing the current instruction and recognising the interrupt, the CPU preserves relevant state, runs the keyboard ISR to obtain the key event, restores state and resumes the interrupted program. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Outline the role of an interrupt and an ISR. [6]

Question 2

Explain where an interrupt can fit into the fetch–decode–execute cycle. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate interrupts compared with repeatedly polling a low-frequency sensor. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An ISR should normally preserve enough execution state to allow the interrupted work to resume.

Use key vocabulary from memory

1. Explain interrupt in your own words.

2. Explain ISR in your own words.

3. Explain context in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.